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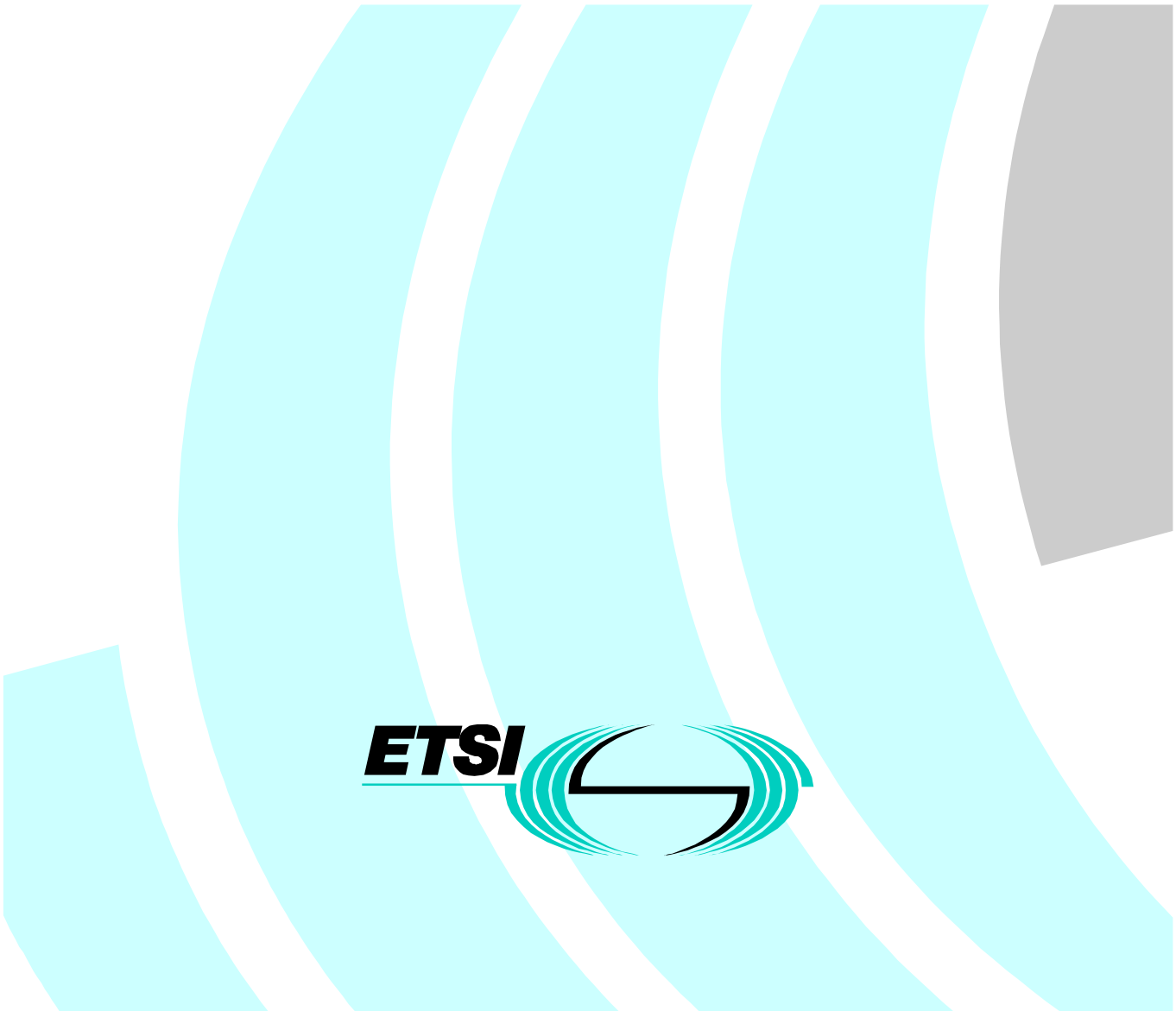
European Standard (Telecommunications series)

Private Integrated Services Network (PISN); Mapping functions for the employment of a circuit mode basic service and the supplementary service user-to-user signalling as a pair of on-demand inter-PINX connections (Mapping/UUS)



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Foreword

This European Standard (Telecommunications series) has been produced by European Computer Manufacturers Association ECMA.

The present document is one of a series of ECMA standards defining mapping functions in exchanges of Private Integrated Services Networks (PISNs) required for the utilization of intervening network scenarios. The series uses the ISDN concepts as developed by ITU-T (formerly CCITT) and is also within the framework of standards for open systems interconnection as defined by ISO.

The present document specifies mapping functions for the type of scenarios where two PINXs are interconnected via on-demand connections via the public ISDN using the supplementary service User-to-User Signalling for carrying signalling information.

The present document is based upon the practical experience of ECMA member companies and the results of their active and continuous participation in the work of ISO/IEC JTC1, ITU-T, ETSI and other international and national standardization bodies. It has been produced under ETSI work item DE/ECMA-00124. It represents a pragmatic and widely based consensus.

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1 Scope

The present document specifies the mapping functions for a pair of on-demand inter-PINX connections using a circuit mode basic service and the supplementary service User-to-User Signalling (service 3) of a public ISDN at the same time.

In order to connect a Private Integrated services Network eXchange (PINX) to another PINX, mapping functions are required to adapt the specific interfaces at the C reference point to the application at the Q reference point. As such, mapping functions provide for physical adaptation to the interface at the C reference point. Mapping functions also provide for the mapping of user channels and signalling information at the Q reference point to the appropriate channels or timeslots at the C reference point.

The C and Q reference points are defined in ISO/IEC 11579-1[3].

At the Q reference point the mappings provide a 64 kbit/s service for user channels and a packet mode service for the signalling channel. Bearer conditioning is outside the scope of the present document, except for providing the layer 2 for the signalling channel at the Q reference point.

Scenario management is outside the scope of the present document.

The present document is applicable to PINXs which can be interconnected to form a PISN and which support signalling protocols at the Q reference point.

2 Conformance

In order to conform to the present document, a PINX shall satisfy the requirements identified in the Implementation Conformance Statement (ICS) proforma in annex A.

3 References

The following documents contain provisions which, through reference in this text, constitute provisions of the present document.

- References are either specific (identified by date of publication, edition number, version number, etc.) or non-specific.
- For a specific reference, subsequent revisions do not apply.
- For a non-specific reference, subsequent revisions do apply.
- A non-specific reference to an ETS shall also be taken to refer to later versions published as an EN with the same number.

[1] EG 201 568 (V1.3.5): "Example 2, using fixed text".

[2] ISO/IEC 11574: "Information technology - Telecommunications and information exchange between systems - Private Integrated Services Network - Circuit-mode 64 kbit/s bearer services - Service description, functional capabilities and information flows (1994)".

[3] ISO/IEC 11579-1: "Information technology - Telecommunications and information exchange between systems - Private integrated services network - Part 1: Reference configuration for PISN Exchanges (PINX) (1994)".

[4] ETS 300 011: "Integrated Services Digital Network (ISDN); Primary rate user-network interface; Layer 1 specification and test principles (1992)".

[5] ETS 300 012: "Integrated Services Digital Network (ISDN); Basic user-network interface; Layer 1 specification and test principles (1992)".

[6] ETS 300 286: "Integrated Services Digital Network (ISDN); User-to-User Signalling (UUS) supplementary service; Digital Subscriber Signalling System No. one (DDS1) protocol; (1995)".

[7] ETS 300 402: "Integrated Services Digital Network (ISDN); Digital Subscriber Signalling System No. one (DSS1) protocol; Data link layer; Application of CCITT Rec. Q.920/I.440 and Q.921/I.441 (1995)".

[8] ETS 300 403: "Integrated Services Digital Network (ISDN); Digital Subscriber Signalling System No. one (DSS1) protocol; Signalling network layer for circuit mode basic call control (1995)".

[9] ETS 300 415: "Private Integrated Services Network (PISN); Terms and definitions (1994)".

[10] CCITT Recommendation I.112: "Vocabulary of terms for ISDNs (1988)".

[11] CCITT Recommendation I.210: "Principles of telecommunication services supported by an ISDN and the means to describe them (1988)".

[12] CCITT Recommendation I.411: "ISDN user-network interfaces - Reference configurations (1988)".

4 Definitions

4.1 External definitions

For the purpose of the present document the following definitions apply:

Basic Service (BS): (CCITT Recommendation I.210 [11])

Call, Basic Call (BC): (ISO/IEC 11582)

Integrated Services Digital Network (ISDN): (CCITT Recommendation I.112 [10])

Private Integrated Services Network (PISN): (ISO/IEC 11579-1[3])

Private Integrated Services Network Exchange (PINX): (ISO/IEC 11579-1[3])

public ISDN: (ETS 300 415 [9])

signalling: (CCITT Recommendation I.112 [10])

Supplementary Service (SS): (ISO/IEC 11582)

4.2 Special definitions

4.2.1 Calling PINX

The PINX which is at the originating interface of the public ISDN InterVening Network (IVN).

4.2.2 Called PINX

The PINX which is at the destination interface of the public ISDN (IVN).

4.2.3 Channel

A means of bi-directional transmission of user or signalling information between two points.

4.2.3.1 D_Q-Channel

A channel used to convey call control information between the Q reference points of two peer PINXs.

4.2.3.2 U_Q-Channel

A channel used to convey user information between the Q reference points of two PINXs.

4.2.4 Inter-PINX Connection

A connection provided by an IVN between two C reference points used to transport inter-PINX information from the PISN control plane and/or the PISN user plane.

4.2.5 Inter-PINX Link

A link between the Q reference points of two PINXs, comprising the totality of signalling transfer and user information transfer means.

5 List of acronyms

DLL	Data Link Layer
DSS1	Digital Subscriber Signalling System No. one
FCS	Field Control Sequence
ICS	Implementation Conformance Statement
IPC	Inter-PINX Connection
IPL	Inter-PINX Link
ISDN	Integrated Services Digital Network
IVN	Intervening Network
PISN	Private Integrated Services Network
PINX	Private Integrated Services Network Exchange
PL	Physical Layer
UUS3	User-to-User Signalling service 3

6 Introduction

The Inter-PINX Connection (IPC) scenario using circuit mode basic services and the supplementary service User-to-User Signalling service 3 (UUS3) as defined for public ISDNs is an on-demand connection scenario.

This scenario creates an Inter-PINX Link (IPL) from the connections established as a result of an ISDN call. The ISDN call is established by specifying the supplementary service UUS3. The signalling connection associated with the ISDN call in conjunction with the UUS3 information transfer capability is used to provide the inter-PINX signalling connection. The circuit mode ISDN connection is used to provide the inter-PINX user connection.

This scenario also includes procedures for sequence control and end-to-end flow control.

Connections of this scenario can be established and released at any time under the control of either PINX. In case of failure, the public ISDN may reject a call establishment request or release an already established call. In case of congestion, the sending PINX may inhibit temporarily the sending of USER INFORMATION signalling messages by means of an internal congestion mechanism.

This scenario provides for the use of up to two pairs of signalling and user information connections per ISDN basic access and up to thirty pairs of signalling and user information connections per ISDN primary rate access (subject to implementation in the public ISDN).

Multiple pairs of IPCs may exist at one or at more interfaces of a PINX. Each pair of IPCs can convey just one IPL. In the case of multiple pairs of IPCs (at one or more interfaces of a PINX), i.e. several calls with UUS3 are established at the same time from one PINX to one or more other PINXs, there is no mapping co-ordination necessary.

Provision of just one number for addressing a Called PINX is sufficient for establishment of any number of pairs of IPCs from a given Calling PINX.

7 Capabilities at the Q reference point

For each instance of the Q reference point:

- one signalling channel (D_Q) for carrying the inter-PINX Layer 3 signalling protocol, and
- one user channel (U_Q)

are provided.

Inter-PINX signalling information is not restricted to call control on this particular U_Q-channel.

For a U_Q-channel the following bearer capability shall be provided:

- information transfer rate: 64 kbit/s;
- other attributes shall be the same as at the C reference point.

NOTE: The provision of bearer conditioning can change these attributes. However, this is outside the scope of the present document.

For a D_Q-channel the following bearer capability shall be provided:

- Transfer mode: packet mode;
- Information transfer capability: unrestricted digital information;
- Information transfer rate: implementation-dependent;
- Other attributes shall be the same as at the C reference point.

The functions to map D_Q and U_Q channels to an IPC at the C reference point are described in clause 8.

8 Mapping functions

The PINX mapping functions shall meet the requirements defined for physical adaptation (see subclause 8.1), channel allocation (see subclause 8.2.1) and bearer conditioning for the D_Q-channel (see subclause 8.2.2).

8.1 Physical adaptation

A PINX shall support at least one of the following physical adaptations.

8.1.1 ISDN primary rate user-network Layer 1 interface

Layer 1 termination shall be in accordance with ETS 300 011 [4].

Timeslots 1 to 15 and 17 to 31 shall be used for up to thirty circuit mode 64 kbit/s IPCs for user information. Timeslot 16 (D-channel) carries signalling information for establishing and clearing the circuit mode 64 kbit/s IPCs and providing, for each one, a packet mode IPC for inter-PINX signalling information through the use of UUS3.

NOTE: UUS3 provides a packet mode service by virtue of allowing the transfer of User-to-User information in USER INFORMATION messages using the call reference of the circuit mode 64 kbit/s IPC. The circuit mode 64 kbit/s IPC will in general have different call references at the calling and called interfaces.

The Layer 2 protocol in timeslot 16 at the C reference point shall conform to:

- ETS 300 402 [7] (Data link).

The Layer 3 protocol in timeslot 16 at the C reference point shall conform to:

- ETS 300 403 [8] (Basic Call control); and
- ETS 300 286 [6] (User-to-User Signalling supplementary service).

8.1.2 ISDN basic user-network Layer 1 interface

Layer 1 termination shall be in accordance with ETS 300 286 [6], excluding application of the

point-to-multipoint mode of operation.

B-channels B1 and B2 of the interface shall be used for up to two circuit mode 64 kbit/s IPCs for user information. The D-channel of the interface carries signalling information for establishing and clearing the circuit mode 64 kbit/s IPCs and providing, for each one, a packet mode IPC for inter-PINX signalling information through the use of UUS3.

NOTE: UUS3 provides a packet mode service by virtue of allowing the transfer of User-to-User information in USER INFORMATION messages using the call reference of the circuit mode 64 kbit/s IPC. The circuit mode 64 kbit/s IPC will in general have different call references at the calling and called interfaces.

The Layer 2 protocol in the D-channel at the C reference point shall conform to:

- ETS 300 402 [7] (Data link).

The Layer 3 protocol in the D-channel at the C reference point shall conform to:

- ETS 300 403 [8] (Basic Call control), and
- ETS 300 286 [6] (User-to-User Signalling supplementary service).

8.2 Mapping matrix

The mapping matrix provides for the mapping of channels at the Q reference point to the IPCs at the C reference point and, in the case of the D_Q-channel, bearer conditioning.

8.2.1 Channel allocation

Per instance of the Q reference point the mapping matrix shall provide for:

- one D_Q-channel. The D_Q-channel shall be mapped onto a packet mode IPC as provided by the physical adaptation function.
- one U_Q-channel. The U_Q-channel number ONE shall be mapped to the corresponding IPC with an information transfer rate of 64 kbit/s as provided by the physical adaptation function.

NOTE: A PINX may additionally support the mapping of a single instance of the Q reference point with more than one U_Q-channel onto a single interface at the C reference point. In this case channel allocation and the numbering of U_Q-channels are implementation matters and not specified in the present document.

8.2.2 Bearer conditioning for the D_Q-channel

The signalling carriage mechanism (Layer 2) on a D_Q-channel shall use a subset of the procedures of ETS 300 402-1 [7] including annex A and ETS 300 402-2 [7] including annex A Data Link Layer (DLL) procedures within the user information field of UUS3. A single DLL frame shall be conveyed in a single User-User information element in a USER INFORMATION message.

The procedures in annex A of ETS 300 402-1 [7] and annex A of ETS 300 402-2 [7] shall apply with the following exceptions:

- The DLL monitor function procedure shall not apply.
- The following DLL commands and responses shall not apply: Un-numbered Information command and Exchange Identification command/response. Following from this, only the following DLL commands and responses shall apply: Information command, Set Asynchronous Balanced Mode Extended command, Disconnect command, Receive Ready command/response, Reject command/response, Receive Not Ready command/response, and Un-numbered Acknowledgement response, Disconnect Mode response and Frame Reject response.
- The DLL frame shall neither contain a starting flag nor a closing flag. The length of a DLL frame can be calculated using the length indicator of the Layer 3 User-User information element in the USER INFORMATION message.
- The DLL frame shall not contain the Field Control Sequence (FCS) field.
- The maximum number of octets in an information field (N 201) shall have the value 125 octets.
- In case of two octet control fields used, the maximum number of N201 is limited to 124 octets.

The service primitives between the DLL and the Physical Layer (PL) shall be interpreted as follows:

- The PL condition "connected" shall apply when the pair of IPCs has been successfully established. The PL condition "disconnected" shall apply when no IPC is available for use.
- The activate-request/indication primitives and deactivate-request/indication primitives shall be mapped to the establishment and clearing of pairs of IPCs.
- The data-request/indication primitives shall be mapped to the request for sending a USER INFORMATION message and to the indication that a USER INFORMATION message has arrived.

9 IPC control functions

9.1 IPC establishment

A call request for establishment of a pair of IPCs is initiated by a scenario management. A request for establishment of a pair of IPCs shall result in a Basic Call (BC) request according to ETS 300 403 [8] and a UUS3 request according to ETS 300 286 [6] from the Calling PINX to the public ISDN.

NOTE 1: The use of UUS3 with a maximum length of only 32 octets user information is outside the scope of the present document.

NOTE 2: Any supplementary service of the intervening ISDN affecting the continuity of the pair of IPCs should not be used.

Selection of appropriate codepoints for the bearer capability, high layer compatibility and low layer compatibility information elements is an implementation matter. However, the selection shall use codepoints that provide a bearer capability suitable for the support of PISN basic services as specified in ISO/IEC 11574 [2].

9.2 IPC clearing

Clearing of an established pair of IPCs shall be performed according to ETS 300 403 [8] and may be initiated by the PINXs on either side of the pair of IPCs. Clearing shall only occur when the related PISN call(s) and/or inter-PINX Layer 3 signalling connection(s) have been cleared already.

NOTE: Whether the pair of IPCs is cleared should be determined taking into account, among other criteria, the delay time for an IPC re-establishment.

10 IPL control functions

10.1 Static pre-conditions for IPL establishment

Each PINX has at least to know:

- that its peer exists and the corresponding PISN number used for IPC establishment;
- that UUS3 scenario applies.

10.2 IPL establishment

IPL establishment is accomplished by establishing a pair of IPCs in accordance with 9.1 and allocating channels at the Q reference point in accordance with subclause 8.2.1.

10.3 Transfer of inter-PINX signalling Layer 2 information

After the pair of IPCs has been established, i.e. after the public ISDN call is in state Active and a UUS3 request has been accepted, the Calling and Called PINXs shall be able to exchange inter-PINX Layer 2 signalling information. The inter-PINX Layer 2 frames shall be embedded in User-User information elements of USER INFORMATION messages. The protocol discriminator of the User-User information element shall be set to value "User-specific protocol".

After answer, in each direction, a burst capability of sending N USER INFORMATION messages is immediately available, where N initially equals the value of the burst parameter X. The value of N shall be decremented by the sending PINX by one for every USER INFORMATION message and incremented by Y at regular intervals of T. The value of N shall be limited to a maximum of X.

The values of T, X and Y are defined in ETS 300 286 [6].

10.4 Use of message segmentation

The maximum size of the user information field inside the User-User information element is 128 octets.

NOTE: The signalling protocol used in the D_Q-channel may need to employ segmentation and reassembly in order to be accommodated in the user information field of a Layer 2 frame conveyed in the User-User information element.

Annex A (normative): Implementation Conformance Statement (ICS) proforma

Notwithstanding the provisions of the copyright clause related to the text of the present document, ETSI grants that users of the present document may freely reproduce the ICS proforma in this annex so that it can be used for its intended purposes and may further publish the completed ICS.

A.1 Introduction

The supplier of an implementation which is claimed to conform to the present document shall complete the Implementation Conformance Statement (ICS) proforma in clause A.3.

A completed ICS proforma is the ICS for the implementation in question. The ICS is a statement of which capabilities and options of the protocol have been implemented. The ICS can have a number of uses, including use:

- by a protocol implementor, as a check-list to reduce the risk of failure to conform to the standard through oversight;
- by the supplier and acquirer (or potential acquirer) of the implementation, as a detailed indication of the capabilities of the implementation, stated relative to the common basis for understanding provided by the standard ICS proforma;
- by the user (or potential user) of the implementation, as a basis for initially checking the possibility of interworking with another implementation (note that, while interworking cannot be guaranteed, failure to interwork can often be predicted from incompatible ICS);
- by a protocol tester, as the basis for selecting appropriate tests against which to asses the claim for conformance of the implementation.

A.2 Instructions for completing the ICS proforma

A.2.1 General structure of the ICS proforma

The ICS proforma is a fixed-format questionnaire divided into subclauses each containing a group of individual items. Each item is identified by an item number, the name of the item (question to be answered) and the reference(s) to the clause(s) that specifies (specify) the item in the main body of the present document.

The "Status" column indicates whether an item is applicable and, if so, whether support is mandatory or optional. The following terms are used:

m	mandatory (the capability is required for conformance to the protocol);
o	optional (the capability is not required for conformance to the protocol, but if the capability is implemented, it is required to conform to the protocol specifications);
o.<n>	optional, but support of at least one of the group of options labelled by the same numeral <n> is required;
x	prohibited;
c.<cond>	conditional requirement, depending on support for the item or items listed in condition <cond>;
<item>:m	simple conditional requirement, the capability being mandatory if item number <item> is supported, otherwise not applicable;

<item>:o simple conditional requirement, the capability being optional if item number <item> is supported, otherwise not applicable.

Answers to the questionnaire items are to be provided either in the "Support" column, by simply marking an answer to indicate a restricted choice (Yes or No) or in the "Not Applicable" column (N/A).

A.2.2 Additional information

Items of additional information allow a supplier to provide further information intended to assist the interpretation of the ICS. It is not intended or expected that a large quantity will be supplied, and a ICS can be considered complete without any such information. Examples might be an outline of the ways in which a (single) implementation can be set up to operate in a variety of environments and configurations.

References to items of additional information may be entered next to any answer in the questionnaire, and may be included in items of exception information.

A.2.3 Exception information

It may occasionally happen that a supplier will wish to answer an item with mandatory or prohibited status (after any conditions have been applied) in a way that conflicts with the indicated requirements. No pre-printed answer will be found in the Support column for this. Instead, the supplier is required to write into the Support column an x.<i> reference to an item of exception information, and to provide the appropriate rationale in the exception item itself.

An implementation for which an exception item is required in this way does not conform to the present document. A possible reason for the situation described above is that a defect in the present document has been reported, a correction for which is expected to change the requirement not met by the implementation.

A.3 ICS proforma for PINX implementation

A.3.1 Implementation identification

Supplier	
Contact point for queries about the ICS	
Implementation name(s) and version(s)	
Other information necessary for full identification, e.g. name(s) and version(s) for machines and/or operating systems; system name(s)	

Only the first three items are required for all implementations; other information may be completed as appropriate in meeting the requirement for full identification.

The terms name and version should be interpreted appropriately to correspond with a supplier’s terminology (e.g. type, series, model).

A.3.2 Implementation summary

Protocol version	1.0
Addenda implemented (if applicable)	
Amendments implemented	
Have any exception items been required (see subclause A.2.3)?	No [ ] Yes [ ] (The answer Yes means that the implementation does not conform to The present document)

Date of Statement	

A.3.3 Procedures

Item	Name of item	Reference	Status	N/A	Support
A1	Support of ISDN primary rate access interface	8.1	o.1		Yes [ ] No [ ]
A2	Support of ISDN basic access interface	8.1	o.1		Yes [ ] No [ ]
A3	Support of ETS 300 402 [7], ETS 300 403 [8] and ETS 300 286 [6]	8.1	m		Yes [ ]
A4	Support of physical adaptation for ISDN primary rate access interface	8.1.1	A1:m	[ ]	Yes [ ]
A5	Support of physical adaptation for ISDN basic access interface	8.1.2	A2:m	[ ]	Yes [ ]
A6	Support of channel allocation	8.2.1	m		Yes [ ]
A7	Support of bearer conditioning for the D_Q-channel	8.2.2	m		Yes [ ]
A8	IPC control functions	9	m		Yes [ ]
A9	IPL control functions	10	m		Yes [ ]

A.3.4 Coding

Item	Name of item	Reference	Status	N/A	Support
B1	Sending of protocol discriminator value 0 of the User-User information element	10.3	m		Yes [ ]
B2	Receipt of protocol discriminator value 0 of the User-User information element	10.3	m		Yes [ ]

History

Document history			
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